

ECL101

Basic Electronics Engineering

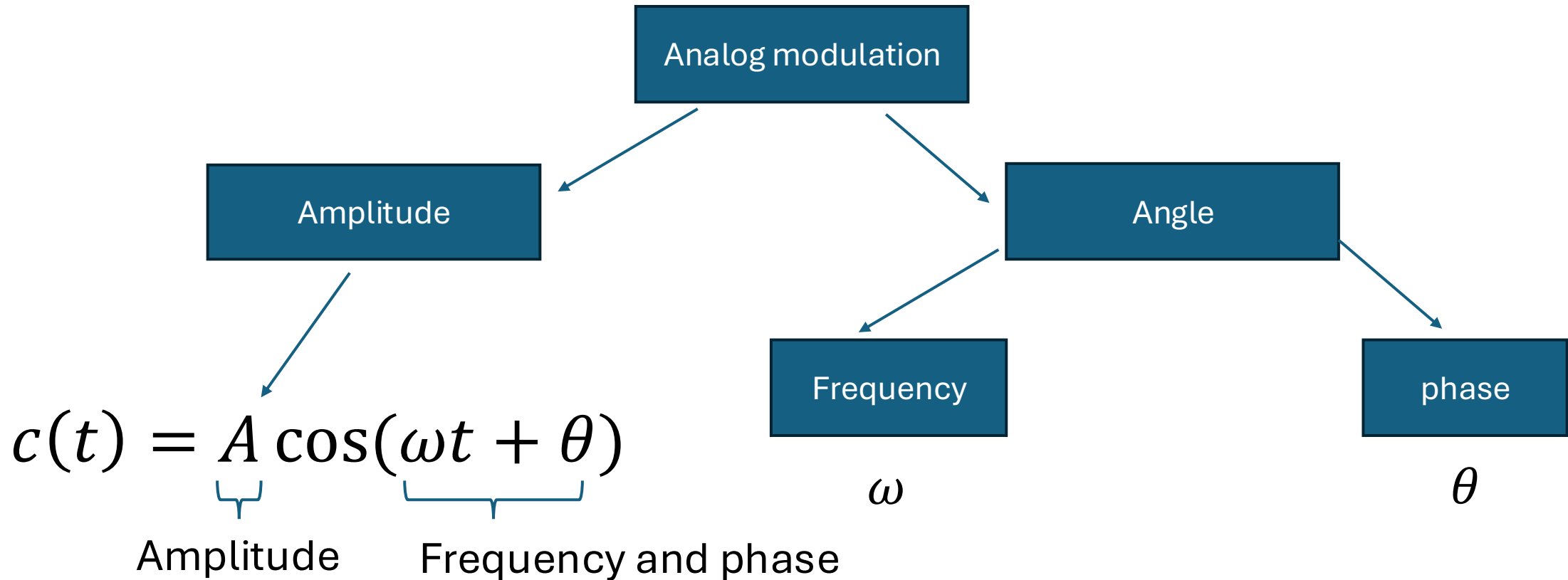
Winter 2025



Dr. Karthikeyan

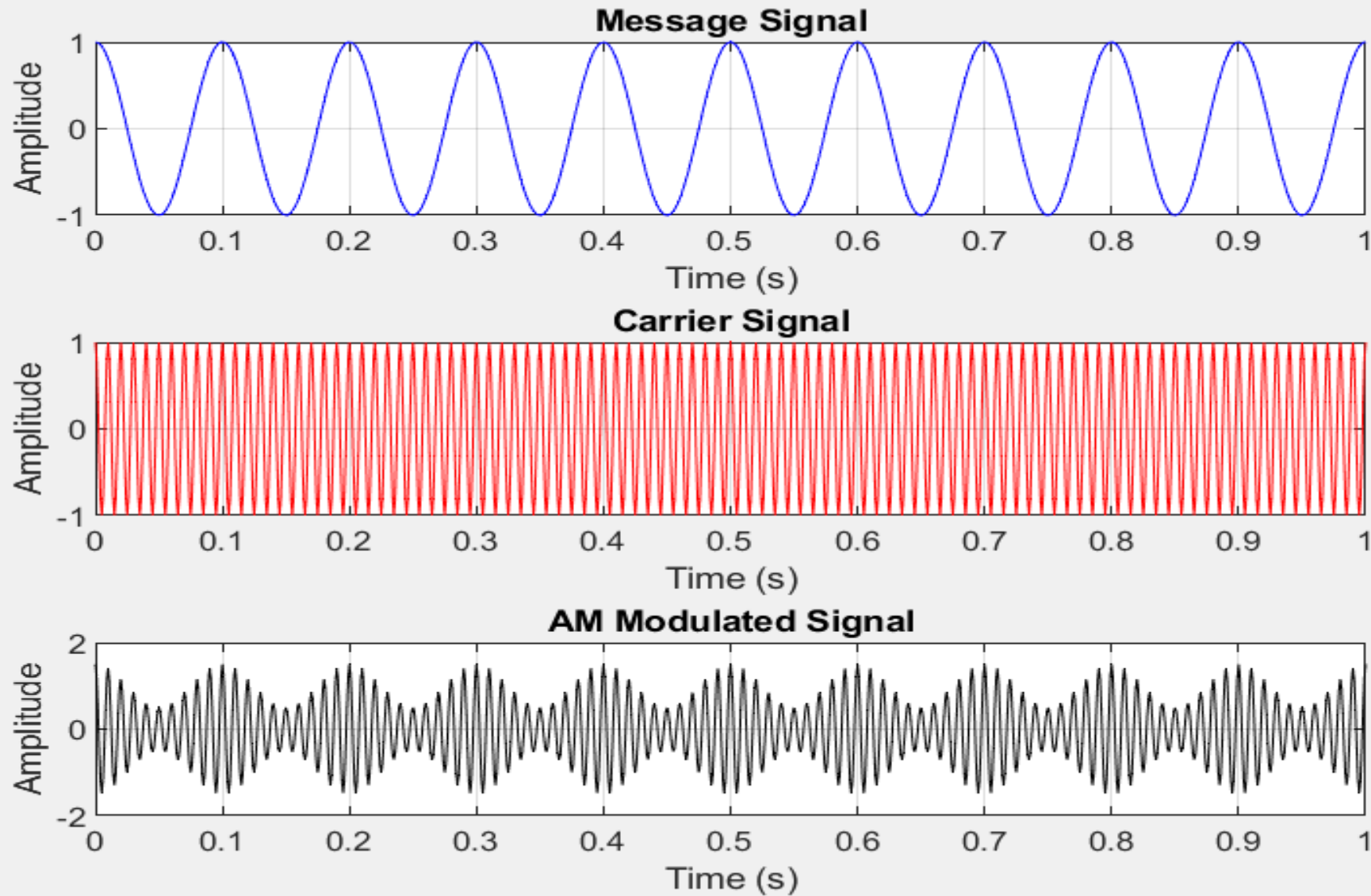
Department of Electronics and Communication Engineering

Analog Modulation



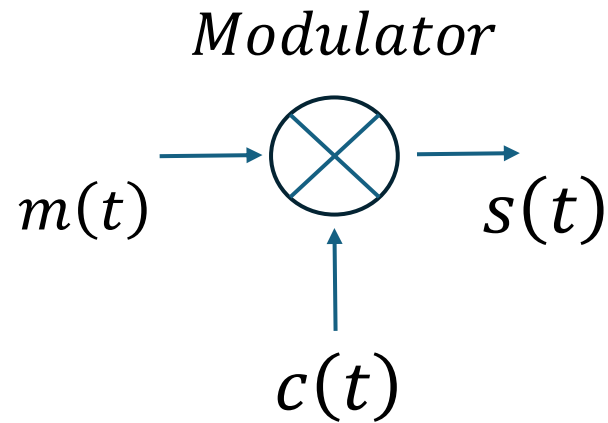
Amplitude Modulation

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Amplitude Modulation

Carrier amplitude (A) is varied in accordance with the message signal $m(t)$



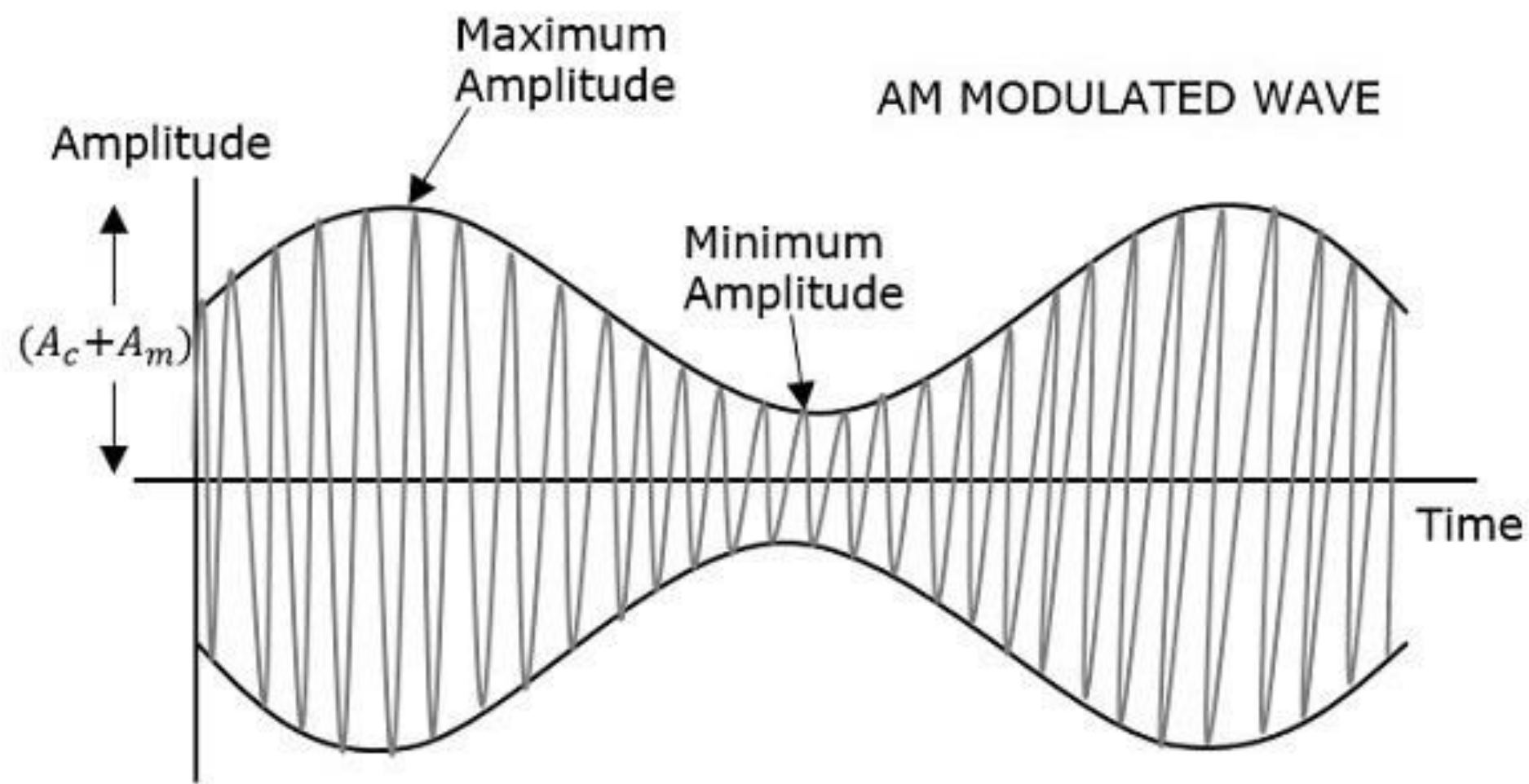
$$s(t) = A_c \cos(\omega_c t) + m(t) \cos(\omega_c t)$$

$$s(t) = [A_c + m(t)] \cos(\omega_c t)$$

$$\text{where } m(t) = A_m \cos(\omega_m t)$$

$$\omega_c \gg \omega_m$$

- ❑ The amplitude of carrier signal is modulated with new amplitude as $[A_c + m(t)]$
- ❑ Hence, the carrier signal is present in the output unmodified along with modulated carrier. Why? **Broadcasting!** There are other schemes where carrier is suppressed.



Amplitude Modulated

Modulating signal be $m(t) = A_m \cos(2\pi f_m t)$

Carrier signal be $c(t) = A_c \cos(2\pi f_c t)$

A_m and A_c are the amplitude of the modulating signal and the carrier signal respectively.

f_m and f_c are the frequency of the modulating signal and the carrier signal respectively.

Amplitude Modulated wave will be

$$s(t) = [A_c + m(t)] \cos(\omega_c t)$$

$$s(t) = [A_c + A_m \cos(2\pi f_m t)] \cos(2\pi f_c t)$$

Modulation Index

It states the level of modulation that a carrier wave undergoes

$$s(t) = [A_c + A_m \cos(2\pi f_m t)] \cos(2\pi f_c t)$$

$$s(t) = A_c \left[1 + \left(\frac{A_m}{A_c} \right) \cos(2\pi f_m t) \right] \cos(2\pi f_c t)$$

$$\Rightarrow s(t) = A_c [1 + \mu \cos(2\pi f_m t)] \cos(2\pi f_c t)$$

$$\mu = \frac{A_m}{A_c}$$

μ is Modulation index

Modulation Index

Maximum amplitude of the modulated wave

$$A_{\max} = A_c + A_m$$

Minimum amplitude of the modulated wave

$$A_{\min} = A_c - A_m$$

$$A_{\max} + A_{\min} = A_c + A_m + A_c - A_m = 2A_c$$

$$\Rightarrow A_c = \frac{A_{\max} + A_{\min}}{2}$$

$$A_{\max} - A_{\min} = A_c + A_m - (A_c - A_m)$$

$$\Rightarrow A_m = \frac{A_{\max} - A_{\min}}{2}$$

$$\frac{A_m}{A_c} = \frac{(A_{\max} - A_{\min})/2}{(A_{\max} + A_{\min})/2}$$

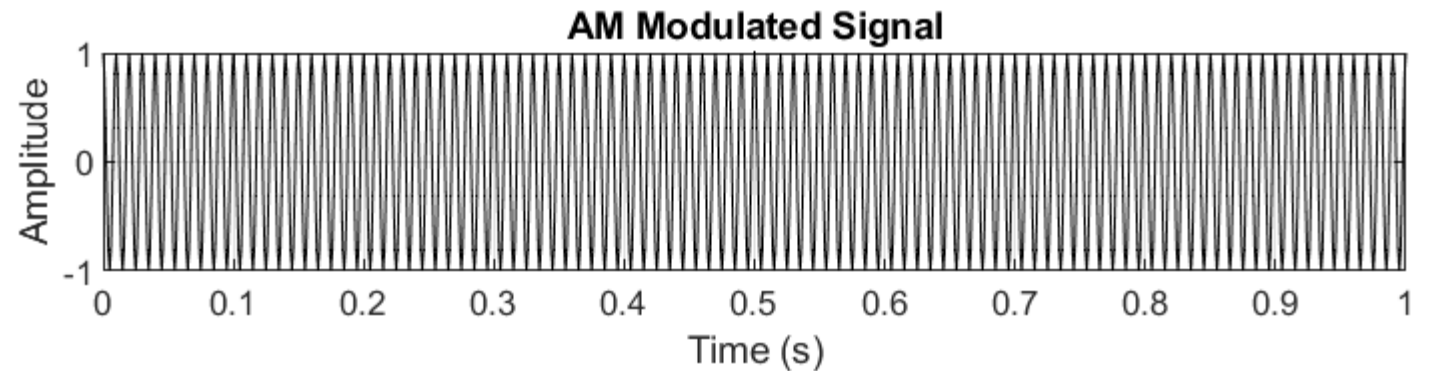
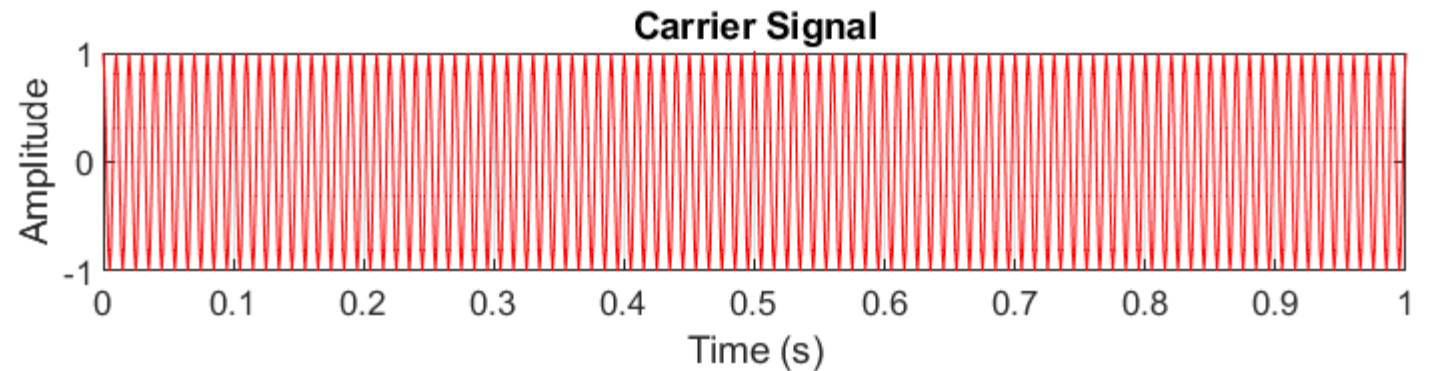
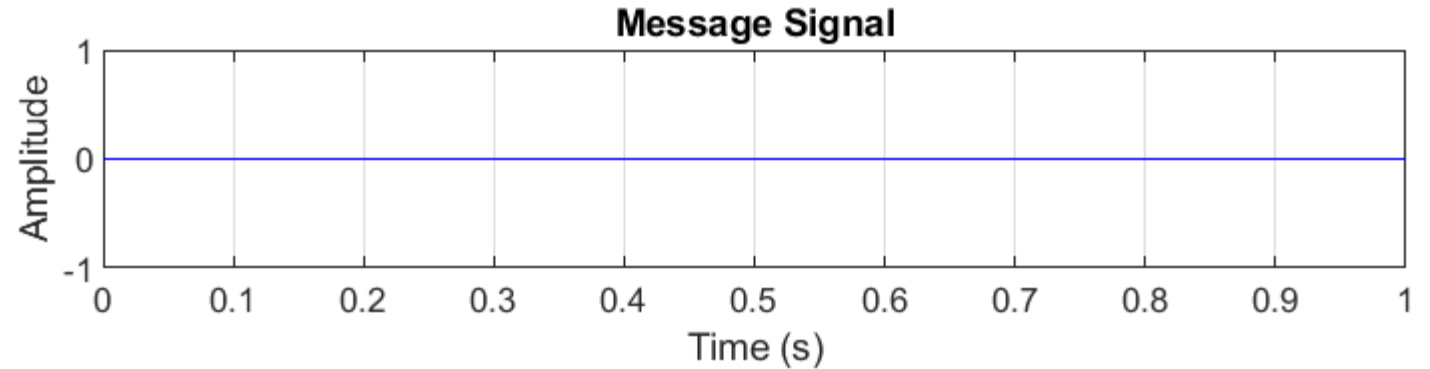
$$\Rightarrow \mu = \frac{A_{\max} - A_{\min}}{A_{\max} + A_{\min}}$$

No message: only carrier

Modulation index

$$\mu = \frac{m_p}{A_c}$$

$$\mu = 0$$

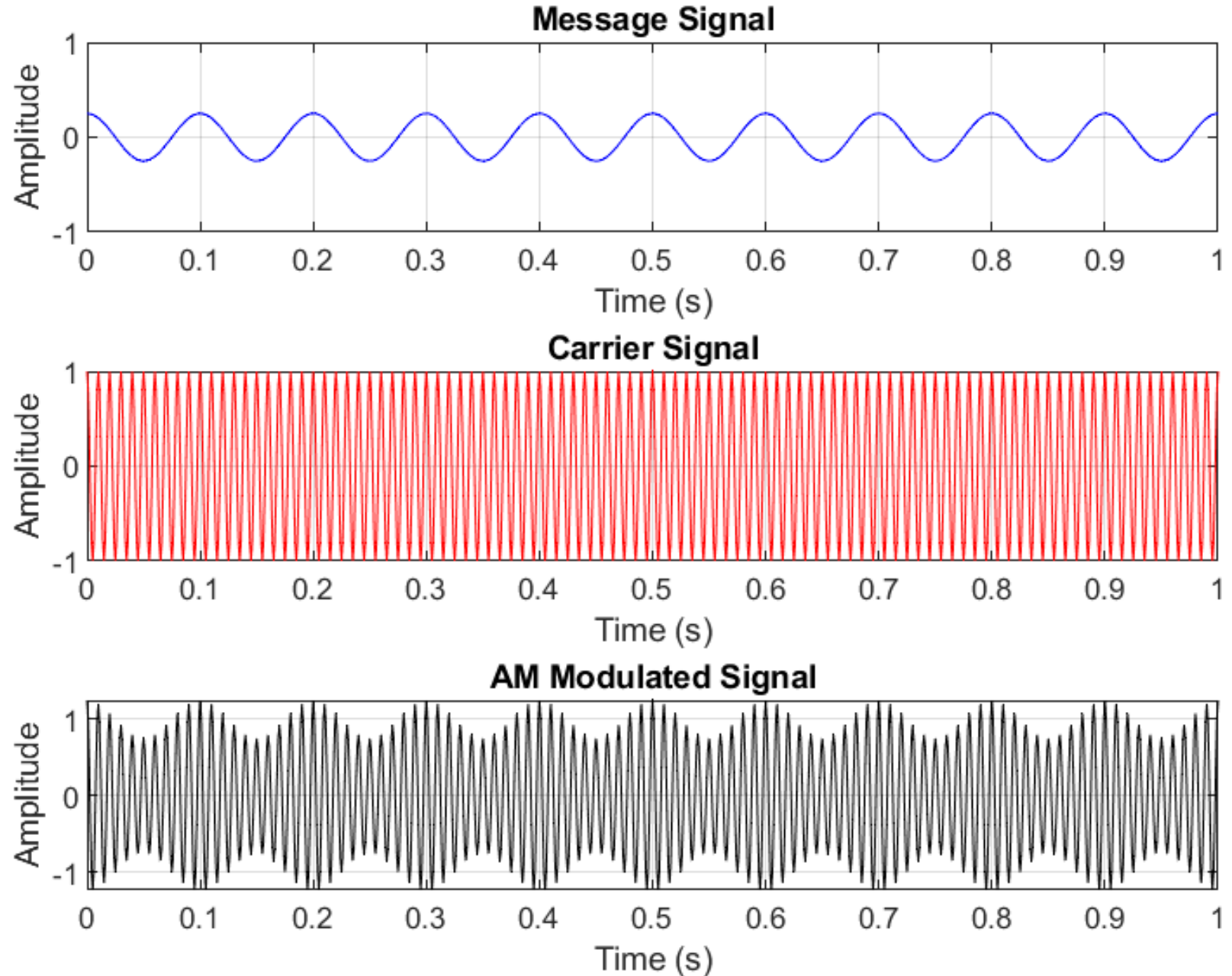


Under modulation

Modulation index

$$\mu = \frac{m_p}{A_c}$$

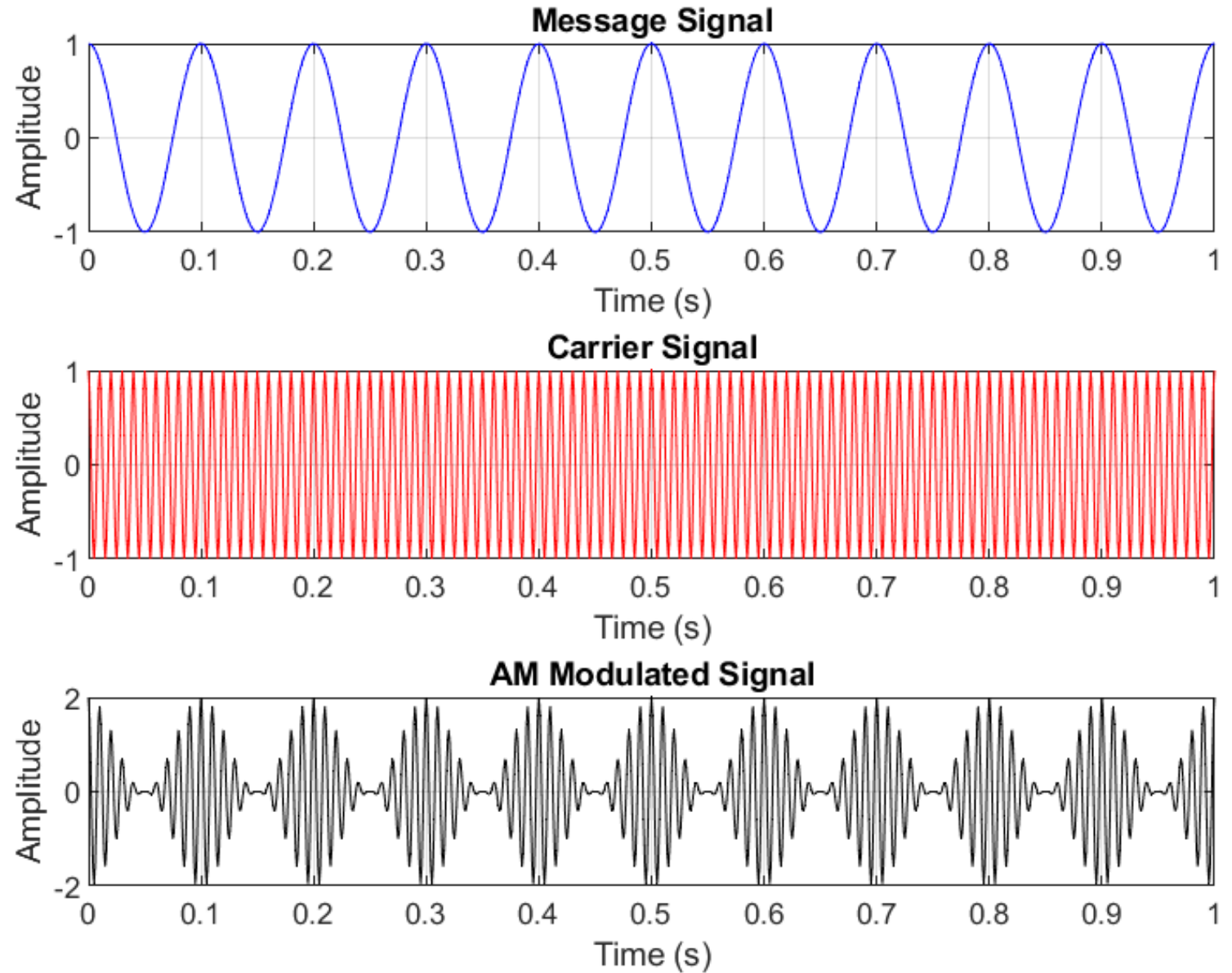
$$\mu = 0.25$$



Critical modulation

$$\mu = \frac{m_p}{A_c}$$

$$\mu = 1$$



Bandwidth of AM Wave

Bandwidth (BW) is difference between the highest and lowest frequencies of the signal.

$$BW = f_{max} - f_{min}$$

$$s(t) = A_c [1 + \mu \cos(2\pi f_m t)] \cos(2\pi f_c t)$$

$$\Rightarrow s(t) = A_c \cos(2\pi f_c t) + A_c \mu \cos(2\pi f_c t) \cos(2\pi f_m t)$$

$$s(t) = A_c \cos(2\pi f_c t) + \frac{A_c \mu}{2} \cos[2\pi (f_c + f_m) t] + \frac{A_c \mu}{2} \cos[2\pi (f_c - f_m) t]$$

amplitude modulated wave has three frequencies. Those are carrier frequency f_c , upper sideband frequency $f_c + f_m$ and lower sideband frequency $f_c - f_m$

Power Calculations of AM Wave

Amplitude modulated wave

$$s(t) = A_c \cos(2\pi f_c t) + \frac{A_c \mu}{2} \cos[2\pi (f_c + f_m) t] + \frac{A_c \mu}{2} \cos[2\pi (f_c - f_m) t]$$

Power of AM wave is equal to the sum of powers of carrier, upper sideband, and lower sideband frequency components.

$$P_t = P_c + P_{USB} + P_{LSB}$$

standard formula for power of cos signal is

$$P = \frac{v_{rms}^2}{R} = \frac{(v_m/\sqrt{2})^2}{2}$$

v_{rms} is the rms value of cos signal.

v_m is the peak value of cos signal.

Power Calculations of AM Wave

Carrier power

$$P_c = \frac{(A_c/\sqrt{2})^2}{R} = \frac{A_c^2}{2R}$$

Upper sideband power

$$P_{USB} = \frac{(A_c\mu/2\sqrt{2})^2}{R} = \frac{A_c^2\mu^2}{8R}$$

lower sideband power

$$P_{LSB} = \frac{A_c^2\mu^2}{8R}$$
$$P_t = \frac{A_c^2}{2R} + \frac{A_c^2\mu^2}{8R} + \frac{A_c^2\mu^2}{8R}$$

Add these three powers in order to get the power of AM wave

$$\Rightarrow P_t = \left(\frac{A_c^2}{2R} \right) \left(1 + \frac{\mu^2}{4} + \frac{\mu^2}{4} \right)$$

$$\Rightarrow P_t = P_c \left(1 + \frac{\mu^2}{2} \right)$$

Problem 1

A modulating signal $m(t) = 10 \cos(2\pi \times 10^3 t)$ is amplitude modulated with a carrier signal $c(t) = 50 \cos(2\pi \times 10^5 t)$. Find the modulation index, the carrier power, and the power required for transmitting AM wave.

Properties of Fourier transform

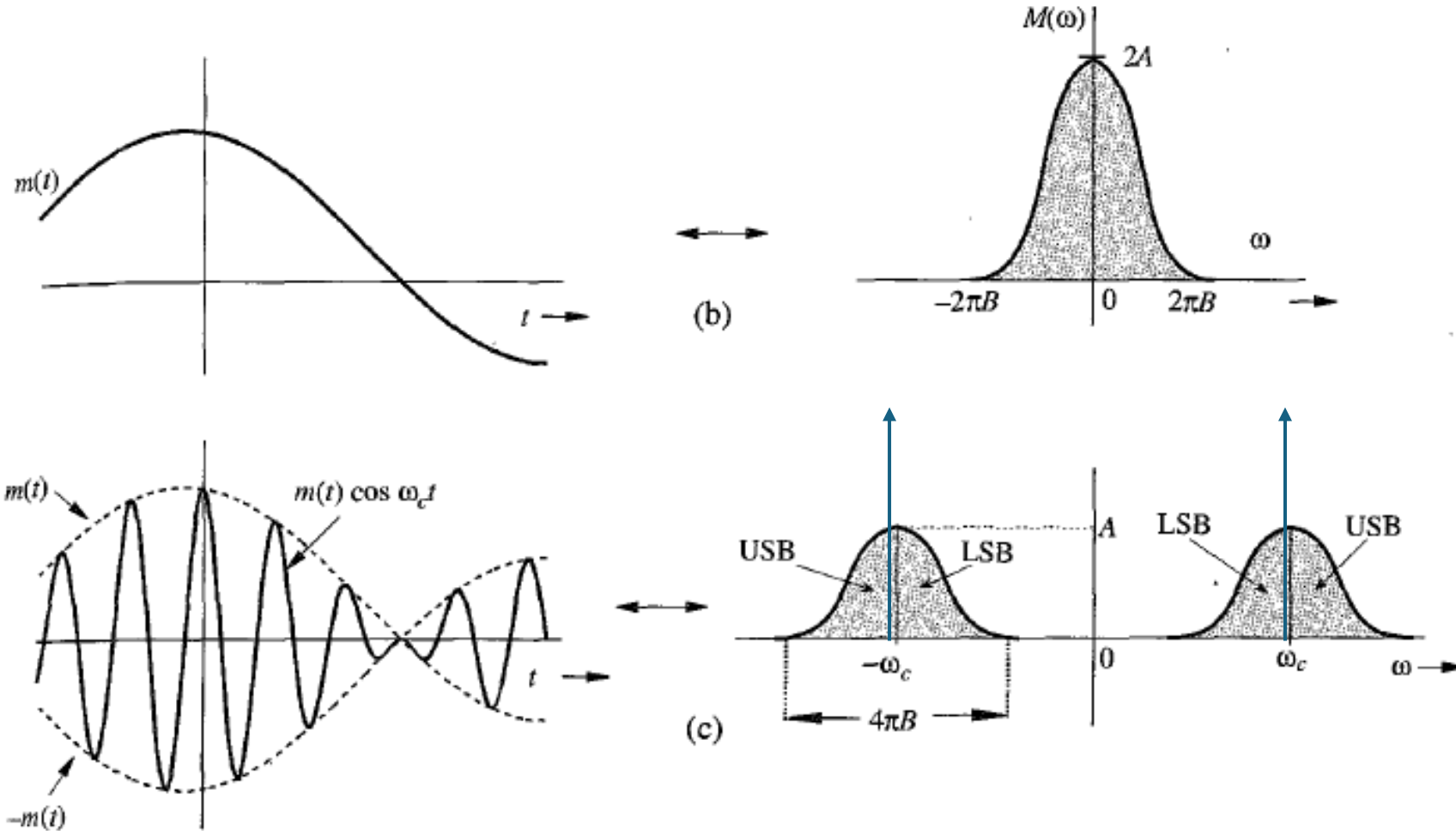
Operation	$g(t)$	$G(f)$
Linearity	$\alpha_1 g_1(t) + \alpha_2 g_2(t)$	$\alpha_1 G_1(f) + \alpha_2 G_2(f)$
Duality	$G(t)$	$g(-f)$
Time scaling	$g(at)$	$\frac{1}{ a } G\left(\frac{f}{a}\right)$
Time shifting	$g(t - t_0)$	$G(f) e^{-j2\pi f t_0}$
Frequency shifting	$g(t) e^{j2\pi f_0 t}$	$G(f - f_0)$
Time convolution	$g_1(t) * g_2(t)$	$G_1(f) G_2(f)$
Frequency convolution	$g_1(t) g_2(t)$	$G_1(f) * G_2(f)$
Time differentiation	$\frac{d^n g(t)}{dt^n}$	$(j2\pi f)^n G(f)$
Time integration	$\int_{-\infty}^t g(x) dx$	$\frac{G(f)}{j2\pi f} + \frac{1}{2} G(0) \delta(f)$

$$\sin(\omega_0 t) = \frac{e^{i\omega_0 t} - e^{-i\omega_0 t}}{2}$$



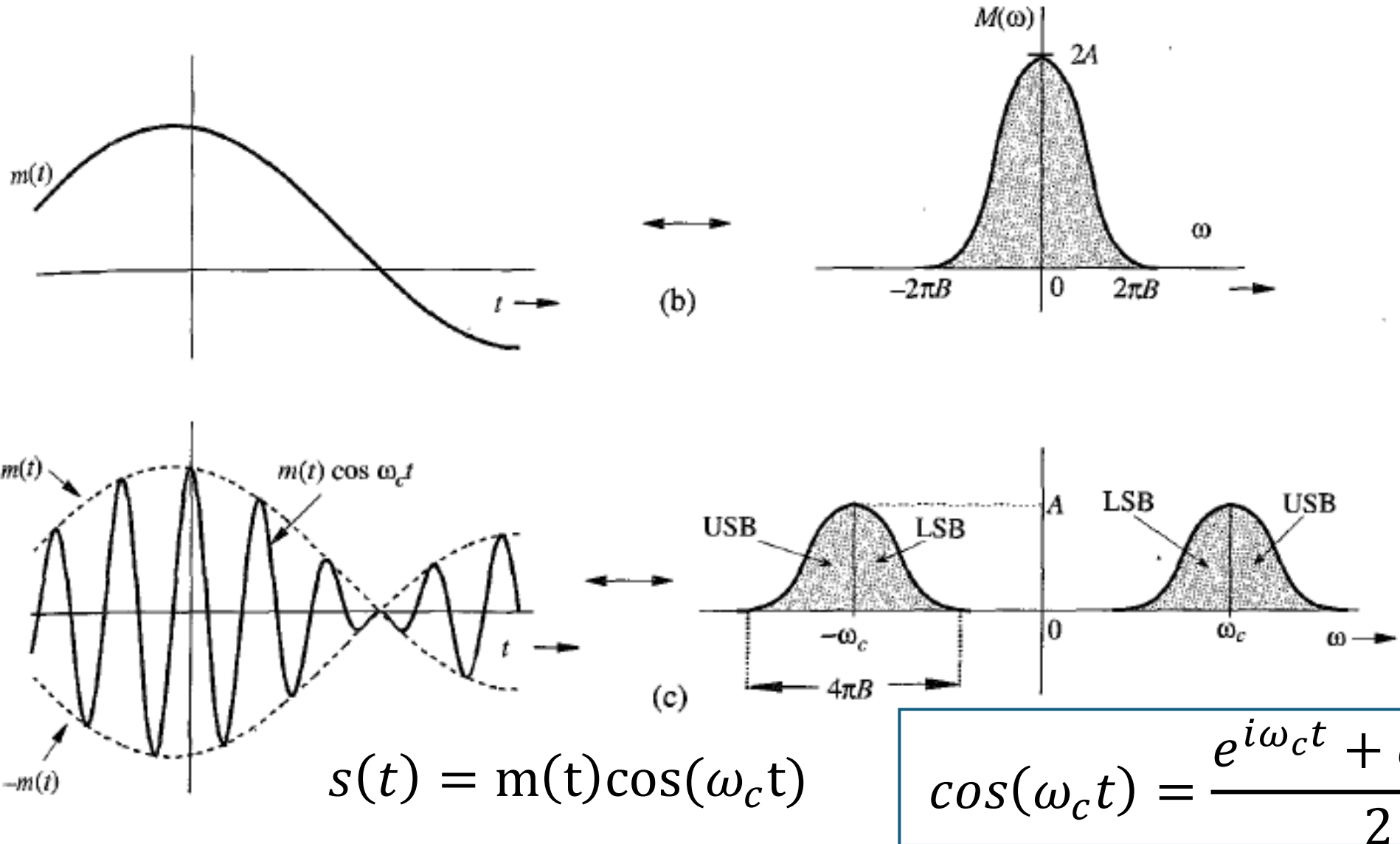
Will shift any spectrum to right and left by amounts $\pm\omega_0$ and scale it by half!

Amplitude Modulation Spectra (AM)



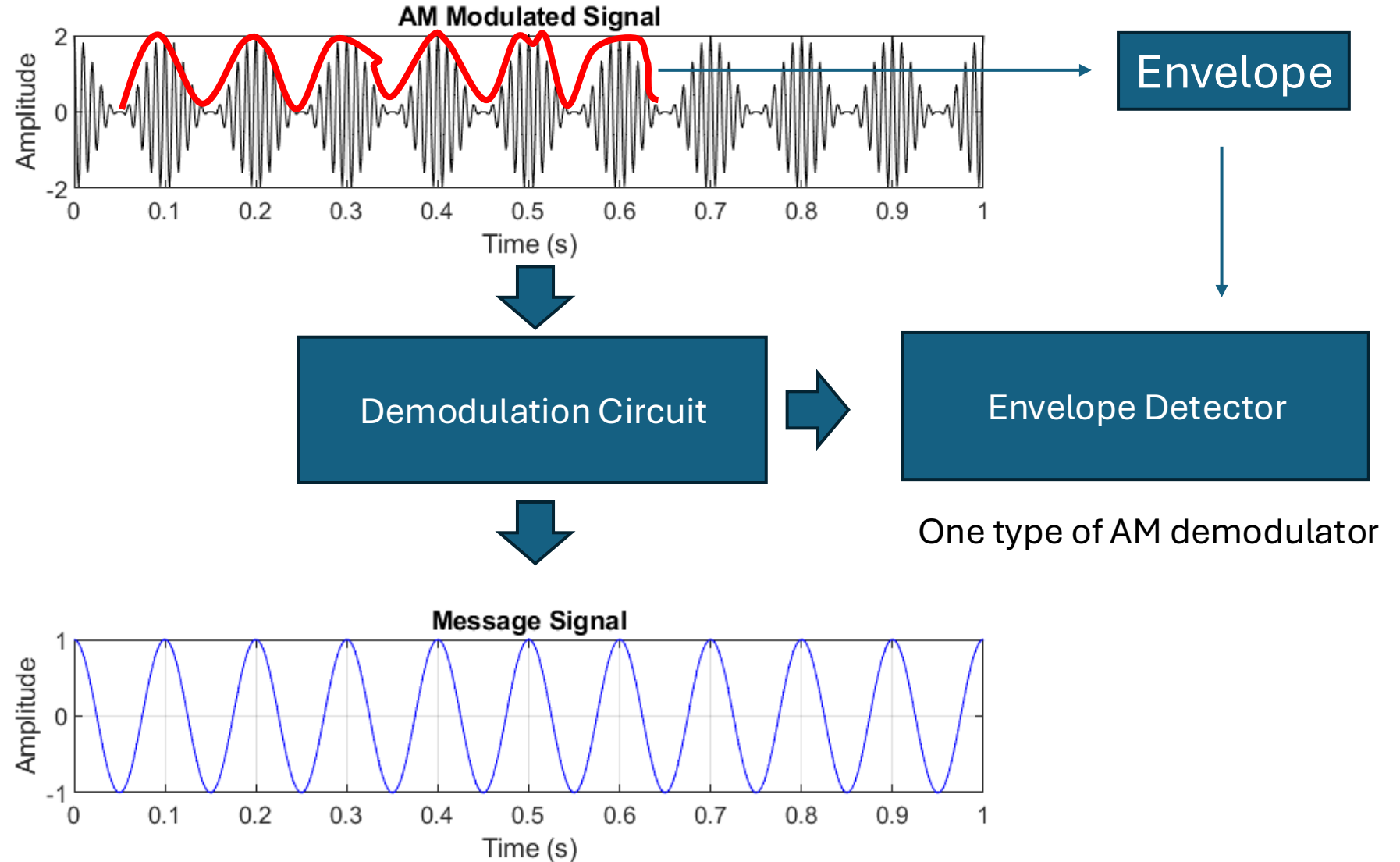
$$s(t) = [A_c + m(t)] \cos(\omega_c t) \quad s(t) = \{A_c\} \cos \omega_c t + m(t) \cos(\omega_c t)$$

Amplitude Modulation Spectra (DSB-SC)



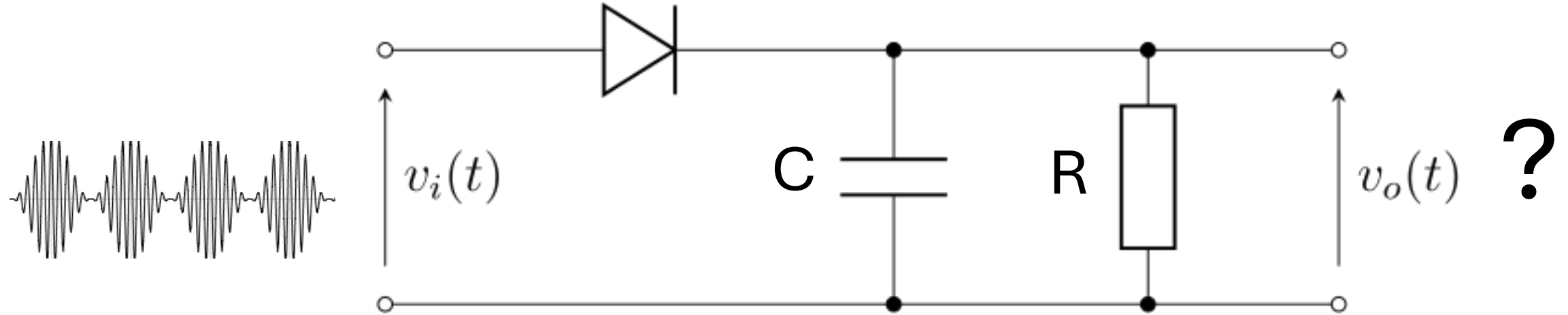
Amplitude Demodulation

Demodulation is the process of retrieving the original message signal from the modulated signal



Envelope Detector

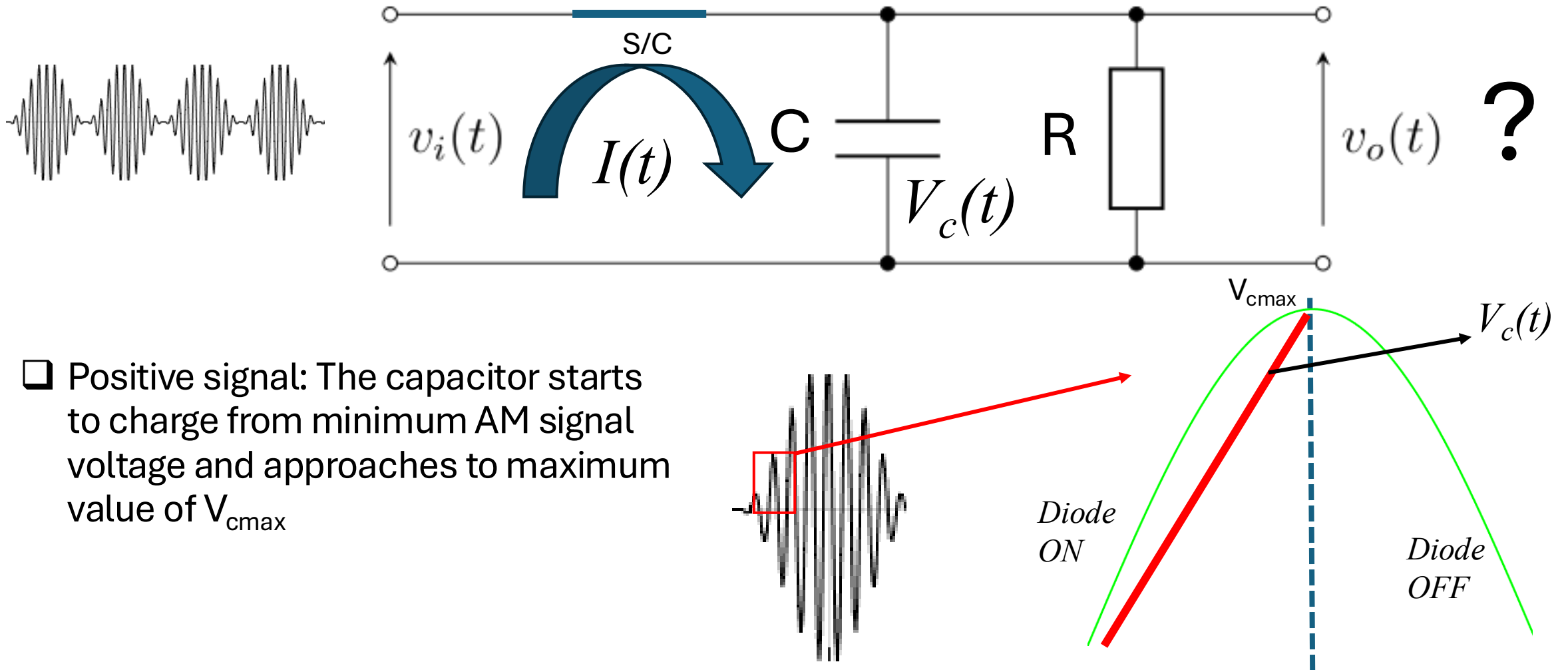
A simple demodulation circuit that uses a PN junction diode, a capacitor and resistor



- ☐ To understand the circuit operation, we need to decimate the operation for positive and negative half cycles of the modulated signal.
- ☐ Circuit behave differently in each half cycle! Let's see how?

Envelope Detector

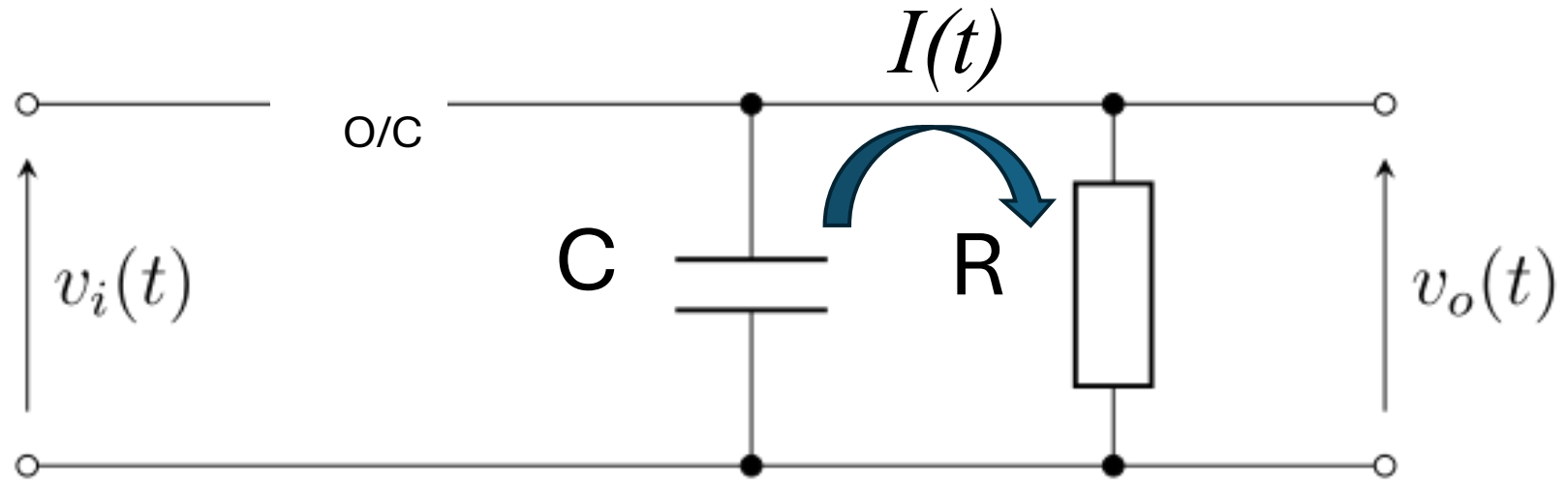
Positive half cycle: Modulated signal is positive and diode conducts (short circuited)



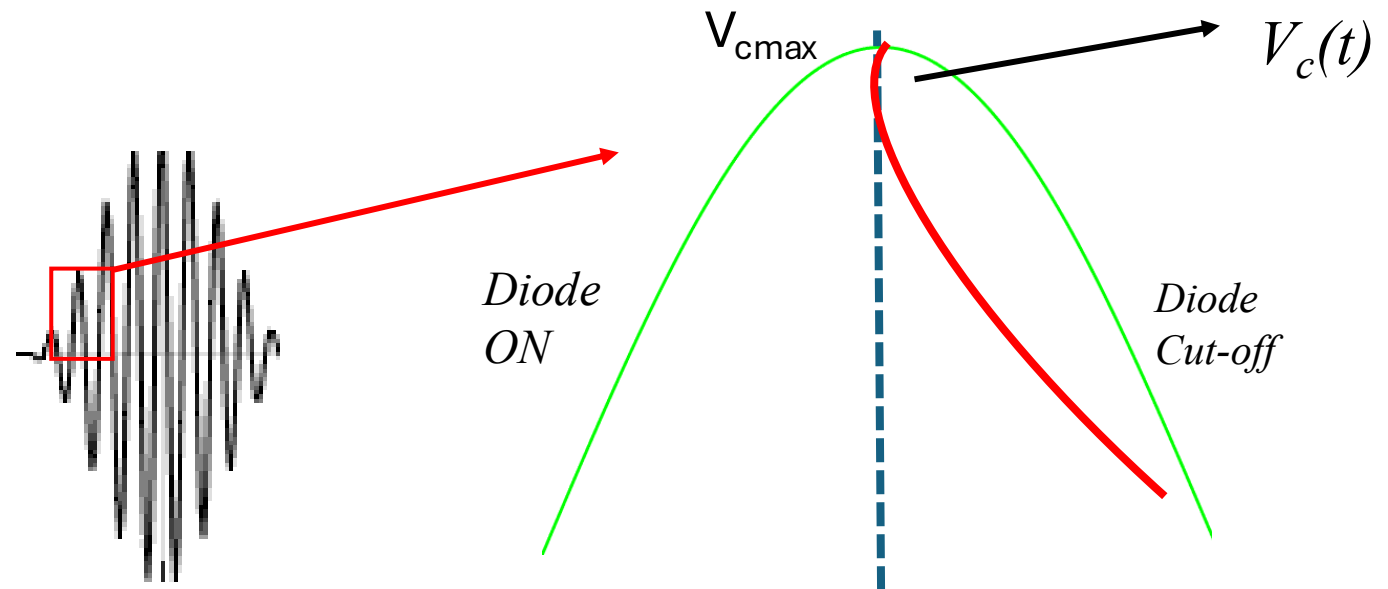
- ❑ Positive signal: The capacitor starts to charge from minimum AM signal voltage and approaches to maximum value of V_{cmax}

Envelope Detector

Negative half cycle: Modulated signal starts falling and diode gets reverse biased (open circuited)



- ❑ Negative signal: The capacitor starts to discharge from peak value until the positive AM signal half-cycle appears again

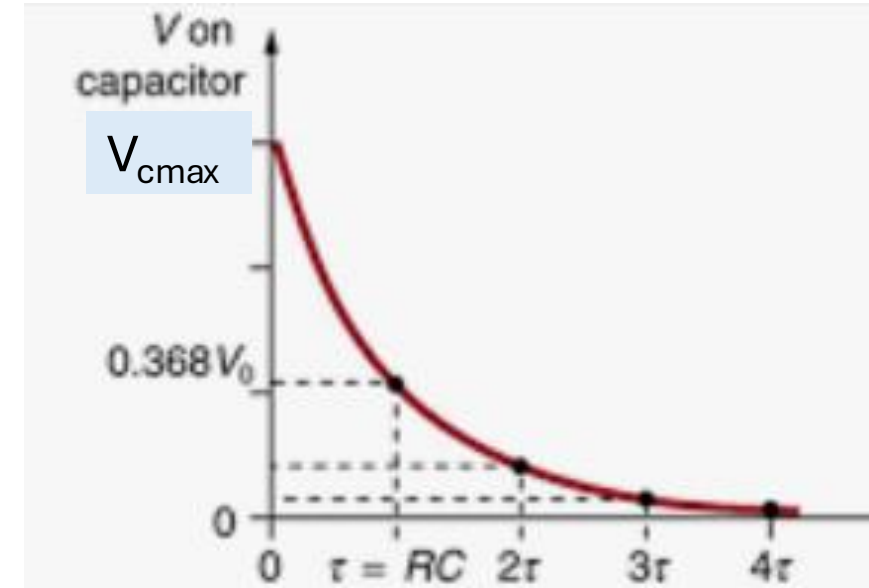
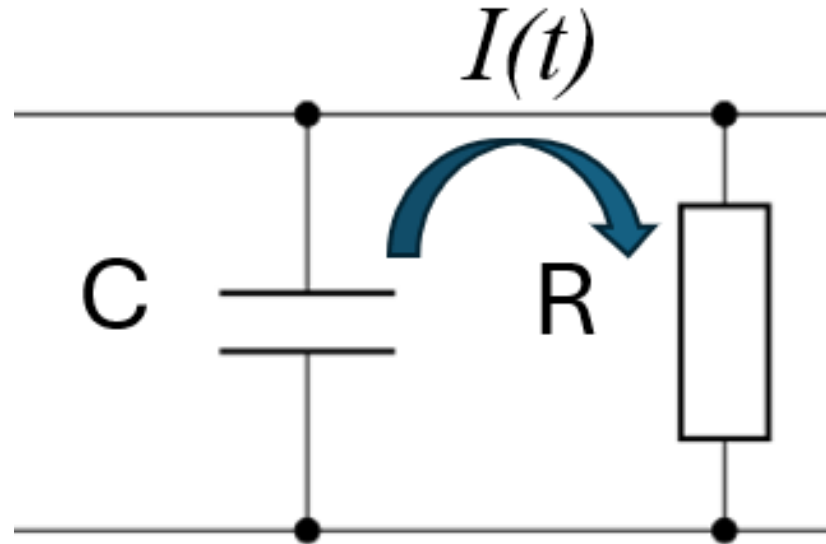


Envelope Detector

- ❑ The rate at which the capacitor charges and discharges depends upon the values of capacitor and resistor
- ❑ The product to RC is called the time constant of the RC circuit and determines the rate of discharge through the resistor R

$$V_c = V_{cmax} e^{\frac{-t}{\tau}}$$

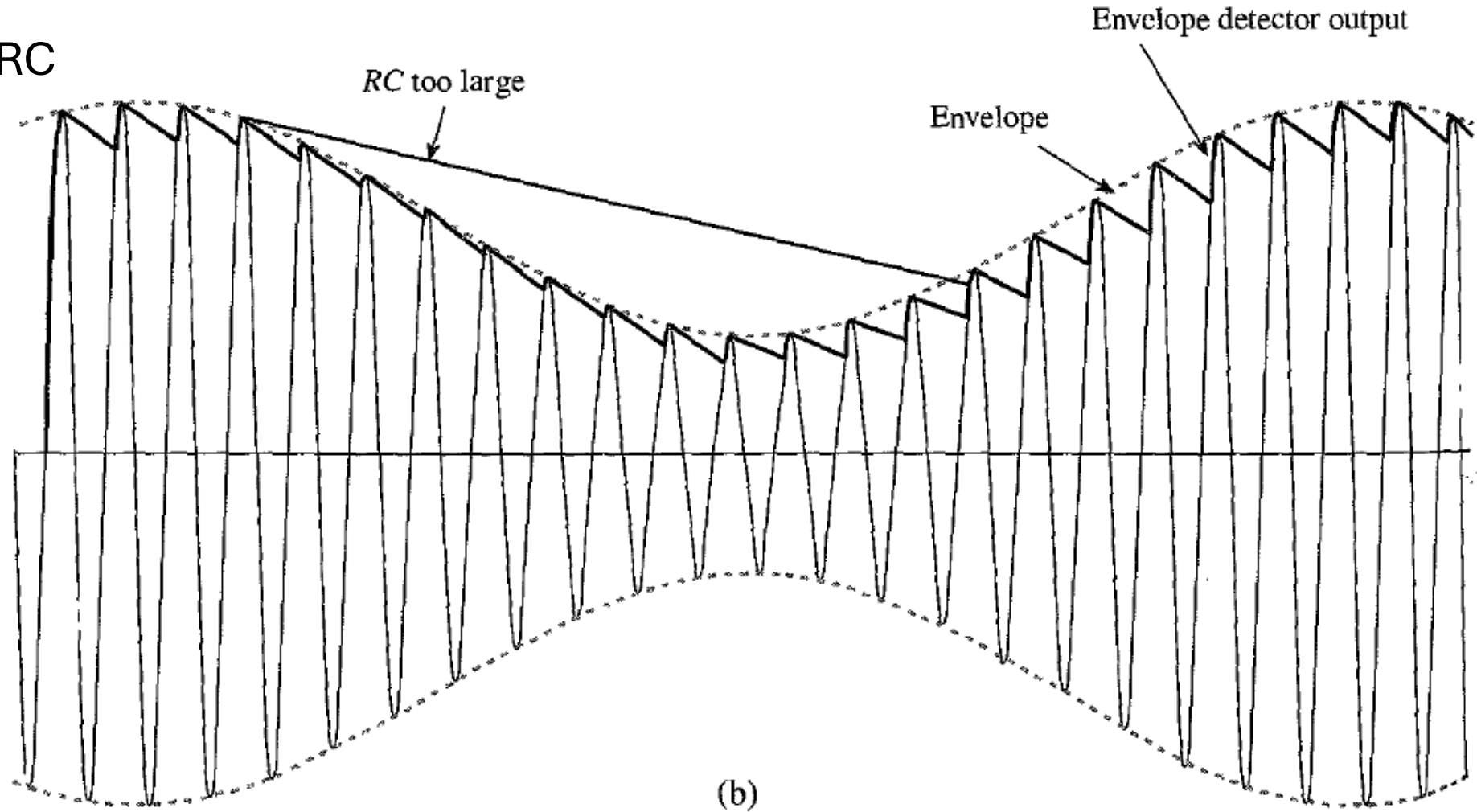
$$\text{where } \tau = RC$$



❑ Therefore, choice of RC becomes very important for proper demodulation!

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❑ Choice of RC

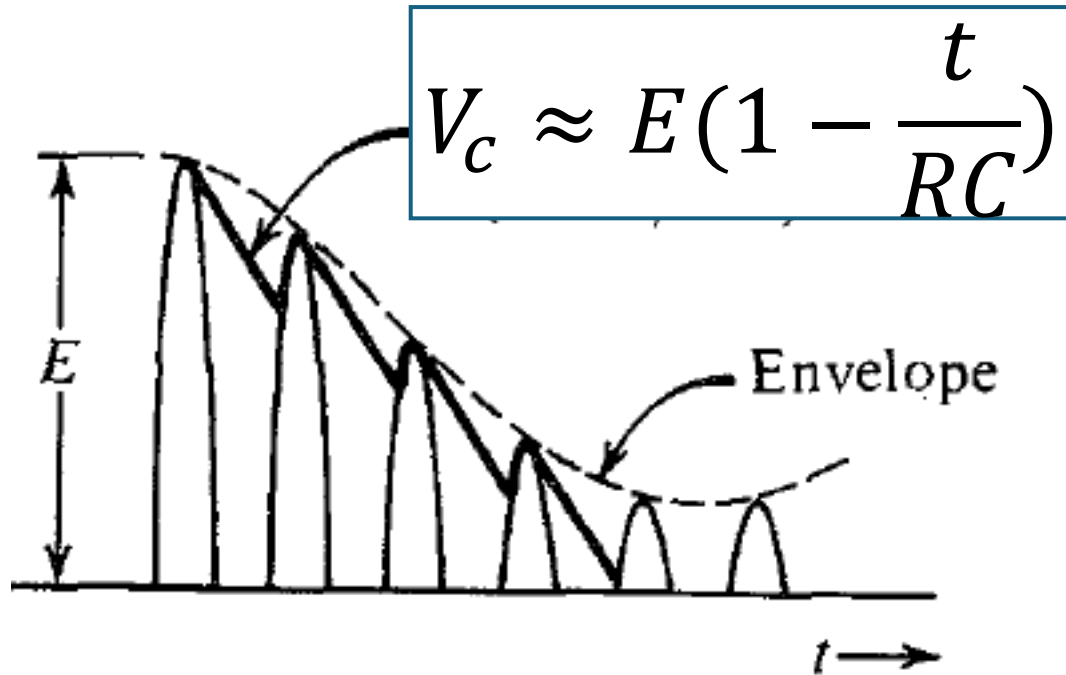


❑ RC should be large compared to $\frac{1}{\omega_c}$

❑ RC should be small compared to $\frac{1}{\omega_m}$
(well defined envelope)

Envelope Detector

Capacitor discharge through RC



$$V_c \approx E(1 - \frac{t}{RC})$$

- The slope of **RC discharge** must be greater than slope of envelope **E(t)**

$$\frac{d}{dt} V_c = \frac{E}{RC} \geq \frac{dE(t)}{dt}$$

$$E(t) = A[1 + \mu \cos \omega_m t]$$

$$\frac{dE(t)}{dt} = -\mu A \omega_m \sin \omega_m t$$

$$RC \leq \frac{1}{\omega_m} \left(\frac{\sqrt{1 - \mu^2}}{\mu} \right) \quad \mu \neq 0$$

- Choice of RC depends on modulation index and should be set as:

